1. The Fundamental Theorem of Calculus states that if F'(x) = f(x) on [a, b], then int_a^b f(x) dx equals:

(A) F(b) + F(a)
(B) F(b) - F(a)
(C) F'(b) - F'(a)
(D) F(a) - F(b)

2. For any continuous function f(x), the definite integral int_a^b f(x) dx is equal to:

(A) int_b^a f(x) dx
(B) - int_b^a f(x) dx
(C) int_0^(b-a) f(x) dx
(D) f(b) - f(a)

3. If f(x) is an odd continuous function on [-a, a], then int_{-a}^a f(x) dx is:

(A) 2 * int_0^a f(x) dx
(B) 0
(C) a * f(a)
(D) -2 * f(0)

4. If f(x) is an even continuous function on [-a, a], then int_{-a}^a f(x) dx is:

(A) 0
(B) 2 * int_0^a f(x) dx
(C) f(a) - f(-a)
(D) int_0^a f(x) dx

5. The value of int_a^b f(x) dx is strictly identical to:

(A) int_a^b f(a + b - x) dx
(B) int_a^b f(a - x) dx
(C) int_a^b f(b - x) dx
(D) int_a^b f(x - a) dx

6. The Gamma function Gamma(n) is defined for n > 0 by the improper integral:

(A) int_0^1 e^(-x) * x^(n-1) dx
(B) int_0^infinity e^(-x) * x^(n-1) dx
(C) int_0^infinity e^(-x^2) dx
(D) int_0^1 x^(n-1) dx

7. The recurrence relation for the Gamma function is Gamma(n + 1) equal to:

(A) Gamma(n)
(B) n * Gamma(n)
(C) (n - 1) * Gamma(n)
(D) Gamma(n) / n

8. For a positive integer n, the value of Gamma(n + 1) is:

(A) n
(B) n!
(C) (n - 1)!
(D) (n + 1)!

9. The value of Gamma(1) is:

(A) 0
(B) 1
(C) pi
(D) Infinity

10. The exact value of Gamma(1/2) is:

(A) 1/2
(B) pi
(C) sqrt(pi)
(D) sqrt(pi) / 2

11. The Beta function Beta(m, n) is defined for m > 0, n > 0 by:

(A) int_0^infinity x^(m-1) * (1-x)^(n-1) dx
(B) int_0^1 x^(m-1) * (1-x)^(n-1) dx
(C) int_0^1 x^m * (1-x)^n dx
(D) int_0^infinity e^(-x) dx

12. The symmetry property of the Beta function states that Beta(m, n) equals:

(A) Beta(n, m)
(B) -Beta(m, n)
(C) 1 / Beta(m, n)
(D) Beta(m+1, n)

13. The fundamental relation connecting the Beta and Gamma functions is Beta(m, n) equal to:

(A) [Gamma(m) + Gamma(n)] / Gamma(m+n)
(B) [Gamma(m) * Gamma(n)] / Gamma(m+n)
(C) Gamma(m+n) / [Gamma(m) * Gamma(n)]
(D) Gamma(m * n)

14. The trigonometric representation of Beta(m, n) is given by:

(A) int_0^(pi/2) sin^(2m-1) theta * cos^(2n-1) theta dtheta
(B) 2 * int_0^(pi/2) sin^(2m-1) theta * cos^(2n-1) theta dtheta
(C) int_0^pi sin^m theta * cos^n theta dtheta
(D) 2 * int_0^pi sin theta dtheta

15. The value of int_0^(pi/2) sin theta dtheta is:

(A) 0
(B) 1
(C) pi / 2
(D) -1

16. The value of int_0^(pi/2) cos theta dtheta is:

(A) 0
(B) 1
(C) pi / 2
(D) 2

17. Wallis' formula evaluates definite integrals of the type:

(A) int_0^infinity e^(-x) dx
(B) int_0^(pi/2) sin^n x dx and int_0^(pi/2) cos^n x dx
(C) int_0^1 x^n dx
(D) int_{-pi}^pi tan x dx

18. The reduction formula for int sin^n x dx expresses the integral in terms of:

(A) int sin^(n-1) x dx
(B) int sin^(n-2) x dx
(C) int cos^n x dx
(D) int sin^(n+1) x dx

19. The reduction formula for int tan^n x dx expresses the integral in terms of:

(A) int tan^(n-1) x dx
(B) int tan^(n-2) x dx
(C) int sec^n x dx
(D) int cot^n x dx

20. The length of arc s of a Cartesian curve y = f(x) from x = a to x = b is given by:

(A) int_a^b sqrt(1 + (dy/dx)) dx
(B) int_a^b sqrt(1 + (dy/dx)^2) dx
(C) int_a^b [1 + (dy/dx)^2] dx
(D) int_a^b sqrt(x^2 + y^2) dx

21. The arc length of a parametric curve x = x(t), y = y(t) from t = t1 to t = t2 is:

(A) int_{t1}^{t2} sqrt((dx/dt)^2 + (dy/dt)^2) dt
(B) int_{t1}^{t2} [(dx/dt) + (dy/dt)] dt
(C) int_{t1}^{t2} sqrt(x^2 + y^2) dt
(D) int_{t1}^{t2} (dy/dx) dt

22. The arc length of a polar curve r = f(theta) from theta = theta1 to theta = theta2 is:

(A) int sqrt(r^2 + (dr/dtheta)) dtheta
(B) int sqrt(r^2 + (dr/dtheta)^2) dtheta
(C) int [r + dr/dtheta] dtheta
(D) int r^2 dtheta

23. The area enclosed between a continuous curve y = f(x) >= 0, the x-axis, and x = a to x = b is:

(A) int_a^b y^2 dx
(B) int_a^b y dx
(C) 2 * pi * int_a^b y dx
(D) int_a^b x * y dx

24. The area enclosed by a polar curve r = f(theta) between theta = alpha and theta = beta is:

(A) int_alpha^beta r dtheta
(B) (1/2) * int_alpha^beta r^2 dtheta
(C) pi * int_alpha^beta r^2 dtheta
(D) 2 * int_alpha^beta r dtheta

25. The volume of the solid generated by revolving the region under y = f(x) from x = a to x = b about the x-axis is:

(A) pi * int_a^b y dx
(B) pi * int_a^b y^2 dx
(C) 2 * pi * int_a^b y^2 dx
(D) pi^2 * int_a^b y dx

26. The volume of the solid generated by revolving y = f(x) about the y-axis (cylindrical shell method) is:

(A) 2 * pi * int_a^b x * y dx
(B) pi * int_a^b x^2 dx
(C) pi * int_a^b y^2 dx
(D) 4 * pi * int_a^b x * y dx

27. The surface area of revolution formed by revolving y = f(x) about the x-axis is:

(A) 2 * pi * int y dx
(B) 2 * pi * int y * sqrt(1 + (y')^2) dx
(C) pi * int y^2 * sqrt(1 + (y')^2) dx
(D) 4 * pi * int y^2 dx

28. In pharmacokinetics, AUC stands for:

(A) Automated Unit Clearance
(B) Area Under the Curve
(C) Active Urinal Concentration
(D) Apparent Uptake Coefficient

29. Total clearance (Cl) of an administered drug dose D is calculated from AUC as:

(A) Cl = D * AUC
(B) Cl = D / AUC
(C) Cl = AUC / D
(D) Cl = sqrt(D * AUC)

30. In bioprocess engineering, cumulative biomass yield over time [0, T] is calculated by:

(A) Derivative of growth rate
(B) Definite integral of production rate over time
(C) Ratio of initial and final cell counts
(D) Logarithmic difference

31. The value of Beta(1, 1) is:

(A) 0
(B) 1
(C) 1/2
(D) 2

32. The value of Gamma(5) is:

(A) 120
(B) 24
(C) 5
(D) 720

33. The value of Gamma(3/2) is:

(A) sqrt(pi)
(B) (1/2) * sqrt(pi)
(C) (3/2) * sqrt(pi)
(D) pi / 2

34. The value of Gamma(5/2) is:

(A) (3/4) * sqrt(pi)
(B) (1/2) * sqrt(pi)
(C) (15/8) * sqrt(pi)
(D) 3 * sqrt(pi)

35. The definite integral int_0^infinity e^(-x^2) dx (Gaussian integral) is equal to:

(A) sqrt(pi)
(B) sqrt(pi) / 2
(C) pi / 2
(D) 1

36. The value of int_0^infinity x^2 * e^(-x) dx is:

(A) 1
(B) 2
(C) 6
(D) 0.5

37. An improper integral of Type 1 has:

(A) Infinite interval of integration
(B) An infinite discontinuity in integrand
(C) Negative integrand
(D) Oscillating limits

38. An improper integral of Type 2 has:

(A) Infinite limits
(B) An integrand that becomes infinite at one or more points in the interval
(C) Complex limits
(D) Only even integrand

39. The integral int_1^infinity (1 / x^p) dx converges if and only if:

(A) p < 1
(B) p > 1
(C) p = 1
(D) p <= 0

40. The integral int_0^1 (1 / x^p) dx converges if and only if:

(A) p > 1
(B) p < 1
(C) p = 1
(D) p >= 2

41. The value of int_0^(pi/2) sin^4 x dx using Wallis' formula is:

(A) 3*pi / 16
(B) pi / 16
(C) 3 / 16
(D) 3*pi / 8

42. The value of int_0^(pi/2) cos^5 x dx using Wallis' formula is:

(A) 8 / 15
(B) 16 / 15
(C) 8*pi / 15
(D) 4 / 15

43. The value of int_0^(pi/2) sin^3 x * cos^4 x dx is:
(A) 2 / 35
(B) 4 / 35
(C) 1 / 35
(D) 2*pi / 35

44. The value of int_0^(pi/2) sin^2 x * cos^2 x dx is:
(A) pi / 8
(B) pi / 16
(C) pi / 32
(D) 1 / 16

45. The value of Beta(2, 3) is:
(A) 1/6
(B) 1/12
(C) 1/20
(D) 1/30

46. The value of Beta(1/2, 1/2) is:
(A) pi
(B) sqrt(pi)
(C) pi / 2
(D) 1

47. Evaluate int_0^infinity x^4 * e^(-2x) dx:
(A) 3/4
(B) 3/8
(C) 3/16
(D) 1/8

48. Evaluate int_0^1 x^3 * (1 - x)^4 dx:
(A) 1/140
(B) 1/280
(C) 1/560
(D) 1/840

49. The perimeter of the astroid x^(2/3) + y^(2/3) = a^(2/3) is:
(A) 4a
(B) 6a
(C) 8a
(D) 2*pi*a

50. The total perimeter of the cardioid r = a*(1 + cos theta) is:
(A) 4a
(B) 6a
(C) 8a
(D) 2*pi*a

51. The area of the loop of the curve y^2 = x^2 * (a - x) is:
(A) 8*a^(5/2) / 15
(B) 4*a^(5/2) / 15
(C) 8*a^3 / 15
(D) 2*a^2 / 3

52. The area of the region enclosed by the cardioid r = a*(1 + cos theta) is:
(A) pi * a^2
(B) 1.5 * pi * a^2
(C) 2 * pi * a^2
(D) 3 * pi * a^2

53. The volume of a sphere of radius R obtained by revolving x^2 + y^2 = R^2 about x-axis is:
(A) 2*pi*R^3
(B) (4/3)*pi*R^3
(C) (2/3)*pi*R^3
(D) pi*R^3

54. The volume of the solid generated by revolving the region bounded by y = x^2 and y = 4 about the y-axis is:
(A) 4*pi
(B) 8*pi
(C) 16*pi
(D) 32*pi

55. Surface area of the sphere formed by revolving the semicircle y = sqrt(R^2 - x^2) about the x-axis is:
(A) 2*pi*R^2
(B) 4*pi*R^2
(C) (4/3)*pi*R^2
(D) pi*R^2

56. Evaluate the definite integral: int_0^pi [x * sin x / (1 + cos^2 x)] dx:
(A) pi^2 / 2
(B) pi^2 / 4
(C) pi / 4
(D) pi^2 / 8

57. Evaluate the definite integral: int_0^(pi/2) ln(sin x) dx:

(A) pi * ln 2
(B) -(pi / 2) * ln 2
(C) (pi / 2) * ln 2
(D) -pi * ln 2

58. Evaluate: int_0^(pi/2) [sqrt(sin x) / (sqrt(sin x) + sqrt(cos x))] dx:

(A) pi
(B) pi / 2
(C) pi / 4
(D) pi / 8

59. Evaluate: int_0^1 [ln(1 + x) / (1 + x^2)] dx:

(A) pi * ln 2 / 4
(B) pi * ln 2 / 8
(C) pi * ln 2 / 2
(D) ln 2 / 4

60. Legendre's duplication formula relates Gamma(m) * Gamma(m + 1/2) to:

(A) Gamma(2m)
(B) [sqrt(pi) / 2^(2m-1)] * Gamma(2m)
(C) [2^(2m-1) / sqrt(pi)] * Gamma(2m)
(D) Gamma(m)^2

61. The relation between Gamma(p) and Gamma(1 - p) for 0 < p < 1 is given by Euler's reflection formula:

(A) pi / sin(p * pi)
(B) sin(p * pi) / pi
(C) sqrt(pi) * sin(p * pi)
(D) pi * cos(p * pi)

62. Evaluate Gamma(1/4) * Gamma(3/4):

(A) pi
(B) pi * sqrt(2)
(C) sqrt(2) / pi
(D) 2 * pi

63. Evaluate int_0^infinity [x^2 / (1 + x^4)] dx:

(A) pi / (2 * sqrt(2))
(B) pi / sqrt(2)
(C) pi / 4
(D) pi / 2

64. Evaluate int_0^infinity [1 / (1 + x^4)] dx:

(A) pi / (2 * sqrt(2))
(B) pi / (4 * sqrt(2))
(C) pi / 2
(D) pi

65. In an intravenous bolus injection, plasma drug concentration follows C(t) = C0 * e^(-k*t). The total AUC from t = 0 to infinity is:

(A) C0 * k
(B) C0 / k
(C) C0 / k^2
(D) k / C0

66. For an oral dose with absorption rate ka and elimination rate ke, C(t) = A*(e^(-ke*t) - e^(-ka*t)). The total AUC from 0 to infinity is:

(A) A * (1/ke - 1/ka)
(B) A * (1/ke + 1/ka)
(C) A / (ke * ka)
(D) A * (ke - ka)

67. The trapezoidal rule for numerical integration approximates int_a^b f(x) dx with step h as:

(A) (h/2) * [(y0 + yn) + 2*(y1 + ... + y_{n-1})]
(B) h * sum y_i
(C) (h/3) * [y0 + 4*y1 + yn]
(D) h * (y0 + yn)

68. Simpson's 1/3 rule requires the total number of subintervals n to be:

(A) Any integer
(B) An even integer
(C) An odd integer
(D) A prime number

69. Simpson's 3/8 rule requires the total number of subintervals n to be:

(A) A multiple of 2
(B) A multiple of 3
(C) A multiple of 4
(D) An odd number

70. The arc length of one complete arch of the cycloid x = a*(theta - sin theta), y = a*(1 - cos theta) is:

(A) 4a
(B) 6a
(C) 8a
(D) 2*pi*a

71. The area enclosed between the parabolas y^2 = 4ax and x^2 = 4ay is:

(A) 16*a^2 / 3
(B) 8*a^2 / 3
(C) 4*a^2 / 3
(D) 32*a^2 / 3

72. The area of the ellipse x^2/a^2 + y^2/b^2 = 1 evaluated by integration is:

(A) 4 * pi * a * b
(B) pi * a * b
(C) 2 * pi * a * b
(D) pi * (a^2 + b^2)

73. The volume of the prolate spheroid formed by revolving the ellipse x^2/a^2 + y^2/b^2 = 1 (a > b) about the x-axis is:

(A) (4/3) * pi * a * b^2
(B) (4/3) * pi * a^2 * b
(C) pi * a * b^2
(D) 2 * pi * a * b^2

74. The volume of the oblate spheroid formed by revolving the ellipse x^2/a^2 + y^2/b^2 = 1 (a > b) about the y-axis is:

(A) (4/3) * pi * a * b^2
(B) (4/3) * pi * a^2 * b
(C) (2/3) * pi * a^2 * b
(D) pi * a^2 * b

75. The derivative of the definite integral with variable limits: d/dx [int_{u(x)}^{v(x)} f(t) dt] is given by:

(A) f(v(x)) - f(u(x))
(B) f(v(x))*v'(x) - f(u(x))*u'(x) (Leibniz Integral Rule)
(C) f'(v(x))*v'(x)
(D) int u'(x) dt

76. Evaluate the reduction integral I_n = int_0^1 x^n * sqrt(1 - x) dx. What is the ratio I_n / I_{n-1}?

(A) 2n / (2n + 3)
(B) n / (n + 1)
(C) 2n / (2n + 1)
(D) (2n - 1) / (2n + 2)

77. Evaluate int_0^(pi/2) sin^6 x * cos^4 x dx using Beta-Gamma functions:

(A) 3*pi / 256
(B) 3*pi / 512
(C) 5*pi / 512
(D) 7*pi / 2048

78. Evaluate int_0^infinity [x^6 * e^(-3x)] dx:

(A) 80 / 243
(B) 80 / 81
(C) 160 / 729
(D) 40 / 243

79. Evaluate int_0^1 [x^5 * (1 - x^3)^10] dx by substituting x^3 = t:

(A) 1 / 396
(B) 1 / 1188
(C) 1 / 594
(D) 1 / 1782

80. Evaluate: int_0^(pi/2) sqrt(tan x) dx:

(A) pi / sqrt(2)
(B) pi / (2 * sqrt(2))
(C) pi * sqrt(2)
(D) 2 * pi

81. Evaluate: int_0^(pi/2) (1 / sqrt(sin x)) dx * int_0^(pi/2) sqrt(sin x) dx:

(A) pi
(B) 2 * pi
(C) pi / 2
(D) pi^2

82. Find the length of the curve y = ln(sec x) from x = 0 to x = pi/3:

(A) ln(2 + sqrt(3))
(B) ln(1 + sqrt(3))
(C) sqrt(3)
(D) 2

83. Find the surface area of the spindle formed by revolving the astroid x = a*cos^3(t), y = a*sin^3(t) about the x-axis:

(A) 6*pi*a^2 / 5
(B) 12*pi*a^2 / 5
(C) 8*pi*a^2 / 5
(D) 4*pi*a^2 / 5

84. The volume of the solid generated by revolving the region bounded by y = e^(-x), y = 0, x = 0, x = infinity about the x-axis is:

(A) pi
(B) pi / 2
(C) 2 * pi
(D) pi / 4

85. The perimeter of the cardioid r = a*(1 - cos theta) from theta = 0 to theta = 2*pi is:

(A) 4a
(B) 6a
(C) 8a
(D) 12a

86. Evaluate: int_0^1 [x^2 / sqrt(1 - x^4)] dx:

(A) sqrt(pi) * Gamma(3/4) / Gamma(1/4)
(B) sqrt(2 * pi^3) / 8
(C) Gamma(3/4) * sqrt(pi) / (4 * Gamma(5/4))
(D) [sqrt(2) / 8] * pi

87. Using Leibniz's rule for differentiation under the integral sign, if F(alpha) = int_0^infinity [e^(-x) * sin(alpha*x) / x] dx, then F(alpha) is:

(A) arctan(alpha)
(B) alpha / (1 + alpha^2)
(C) ln(1 + alpha^2)
(D) e^(-alpha)

88. Evaluate the Frullani integral int_0^infinity [(e^(-a*x) - e^(-b*x)) / x] dx (where a, b > 0):

(A) ln(b / a)
(B) ln(a / b)
(C) b - a
(D) (b - a) / (a * b)

89. Evaluate the Frullani integral int_0^infinity [(cos(a*x) - cos(b*x)) / x] dx (where a, b > 0):

(A) ln(b / a)
(B) 0
(C) ln(a / b)
(D) pi / 2

90. In a bioprocess fed-batch culture, product concentration P(t) increases as dP/dt = q_p * X(t). If X(t) = X0 * e^(mu*t), total product formed P(T) - P(0) is:

(A) (q_p * X0 / mu) * (e^(mu*T) - 1)
(B) q_p * X0 * T
(C) (q_p * X0 / mu) * e^(mu*T)
(D) q_p * mu * X0 * T^2

91. The area of the surface formed by revolving one loop of the lemniscate r^2 = a^2 * cos(2*theta) about the polar axis is:

(A) 2 * pi * a^2 * (2 - sqrt(2))
(B) pi * a^2 * sqrt(2)
(C) 4 * pi * a^2
(D) 2 * pi * a^2

92. Evaluate: int_0^(pi/4) ln(1 + tan theta) dtheta:

(A) (pi / 4) * ln 2
(B) (pi / 8) * ln 2
(C) (pi / 2) * ln 2
(D) (pi / 16) * ln 2

93. Evaluate: int_0^1 [x^(p-1) + x^(q-1)] / (1 + x)^(p+q) dx:

(A) Beta(p, q)
(B) 2 * Beta(p, q)
(C) Beta(p, q) / 2
(D) Gamma(p+q)

94. The length of the catenary y = c * cosh(x / c) from x = 0 to x = a is:

(A) c * sinh(a / c)
(B) c * cosh(a / c)
(C) a * sinh(c / a)
(D) c * tanh(a / c)

95. Find the volume of the solid generated by revolving the cissoid y^2*(2a - x) = x^3 about its asymptote x = 2a:

(A) 2 * pi^2 * a^3
(B) pi^2 * a^3
(C) 4 * pi^2 * a^3
(D) (4/3) * pi^2 * a^3

96. The centroid of the area bounded by the parabola y^2 = 4ax and its latus rectum x = a has x-coordinate x_bar:

(A) 3a / 5
(B) 2a / 5
(C) a / 2
(D) 4a / 5

97. The moment of inertia of a uniform circular disc of mass M and radius R about its perpendicular central axis is:

(A) M * R^2
(B) (1/2) * M * R^2
(C) (1/4) * M * R^2
(D) (2/5) * M * R^2

98. A drug is eliminated according to two-compartment kinetics: C(t) = 40*e^(-2t) + 10*e^(-0.1t) (ug/mL). The total AUC(0 to infinity) is:

(A) 20 + 100 = 120 ug*h/mL
(B) 40 + 10 = 50 ug*h/mL
(C) 200 ug*h/mL
(D) 80 ug*h/mL

99. Evaluate the integral int_0^infinity [x / (1 + x^6)] dx:

(A) pi / (3 * sqrt(3))
(B) pi / 6
(C) pi / (2 * sqrt(3))
(D) pi / 3

100. Evaluate the integral int_0^infinity [x^4 / (1 + x^10)] dx:

(A) pi / 10
(B) pi / 5
(C) pi / 20
(D) 1 / 10

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. State and prove the Fundamental Theorem of Calculus (Parts 1 and 2) using Riemann sums and the Mean Value Theorem for integrals.

Q2. Derive the reduction formula for I_n = int sin^n x dx. Hence, deduce Wallis' Formula for int_0^(pi/2) sin^n x dx for both even and odd positive integers n.

Q3. Derive the reduction formula for I_n = int cos^n x dx. Using Wallis' formula, evaluate: (i) int_0^(pi/2) cos^7 x dx, and (ii) int_0^(pi/2) cos^8 x dx.

Q4. Derive the reduction formula for I_{m,n} = int sin^m x * cos^n x dx. Hence, evaluate the definite integral: int_0^(pi/2) sin^6 x * cos^4 x dx.

Q5. Derive the reduction formulas for: (i) I_n = int tan^n x dx, and (ii) I_n = int sec^n x dx. Using these, evaluate int_0^(pi/4) tan^5 x dx and int sec^3 x dx.

Q6. Derive the reduction formula for I_n = int x^n * e^(a*x) dx. Using this formula, evaluate the definite integral: int_0^1 x^4 * e^(2x) dx.

Q7. Derive the reduction formula for I_{m,n} = int x^m * (ln x)^n dx. Hence evaluate the definite integral: int_0^1 x^3 * (ln x)^2 dx.

Q8. Define the Gamma function Gamma(n). Prove that the integral converges for all n > 0. Establish the recurrence relation Gamma(n + 1) = n * Gamma(n) and deduce that Gamma(n + 1) = n! for non-negative integers.

Q9. Prove that Gamma(1/2) = sqrt(pi) by transforming the product of two Gaussian integrals into polar coordinates.

Q10. Define the Beta function Beta(m, n). Prove: (i) Beta(m, n) = Beta(n, m) (symmetry), and (ii) Beta(m, n) = 2 * int_0^(pi/2) sin^(2m-1) theta * cos^(2n-1) theta dtheta (trigonometric form).

Q11. Prove the fundamental relation connecting Beta and Gamma functions: Beta(m, n) = [Gamma(m) * Gamma(n)] / Gamma(m + n).

Q12. State and prove Legendre's Duplication Formula for the Gamma function: Gamma(m) * Gamma(m + 1/2) = [sqrt(pi) / 2^(2m - 1)] * Gamma(2m).

Q13. State Euler's Reflection Formula for the Gamma function. Use it to evaluate: (i) Gamma(1/4) * Gamma(3/4), and (ii) int_0^infinity [1 / (1 + x^4)] dx.

Q14. Evaluate the definite integrals using Beta and Gamma functions: (i) int_0^1 x^4 * (1 - sqrt(x))^5 dx, and (ii) int_0^(pi/2) sqrt(tan theta) dtheta.

Q15. Prove that int_0^(pi/2) ln(sin x) dx = - (pi / 2) * ln 2 using properties of definite integrals.

Q16. Prove that int_0^pi [x * sin x / (1 + cos^2 x)] dx = pi^2 / 4.

Q17. Derive the Cartesian formula for arc length (rectification): s = int_a^b sqrt(1 + (dy/dx)^2) dx. Find the total perimeter of the astroid: x^(2/3) + y^(2/3) = a^(2/3).

Q18. Find the arc length of one complete arch of the cycloid: x = a * (theta - sin theta), y = a * (1 - cos theta) (where theta ranges from 0 to 2*pi).

Q19. Derive the polar arc length formula: s = int_alpha^beta sqrt(r^2 + (dr/dtheta)^2) dtheta. Hence, compute the total perimeter of the cardioid r = a * (1 + cos theta).

Q20. Find the arc length of the catenary y = c * cosh(x / c) between x = -a and x = a.

Q21. Find the area bounded between the two intersecting parabolas y^2 = 4ax and x^2 = 4ay (where a > 0).

Q22. Derive the polar area formula A = (1/2) * int_alpha^beta r^2 dtheta. Hence, compute the area of the entire region enclosed by the cardioid r = a * (1 + cos theta).

Q23. Derive the Disk Method for the volume of a solid of revolution. Use it to prove that the volume of a sphere of radius R is (4/3) * pi * R^3.

Q24. Derive the Cylindrical Shell Method for the volume of revolution about the y-axis: V = 2*pi * int_a^b x * y dx. Use this method to find the volume generated by revolving y = 4 - x^2 (for 0 <= x <= 2) about the y-axis.

Q25. Derive the formula for the surface area of a solid of revolution about the x-axis: S = 2*pi * int y * sqrt(1 + (y')^2) dx. Use it to prove that the surface area of a sphere of radius R is 4 * pi * R^2.

Q26. State Leibniz's Rule for differentiation under the integral sign with variable limits. Use it to evaluate the Frullani integral: int_0^infinity [(e^(-a*x) - e^(-b*x)) / x] dx (where a, b > 0).

Q27. Derive Simpson's 1/3 Rule for numerical integration from the Lagrange quadratic interpolating polynomial. Analyze its order of accuracy and error term.

Q28. In biotechnology and pharmacokinetics, define Area Under the Curve (AUC). Calculate total AUC(0 to infinity) and clearance Cl for a two-compartment intravenous drug bolus C(t) = 50*e^(-1.5*t) + 20*e^(-0.2*t) (mg/L, t in hours) with an initial dose D = 500 mg.

Q29. In a fed-batch bioprocess, a recombinant protein product P is produced with rate dP/dt = q_p * X(t) * V(t). Given biomass X(t) = X0 * e^(mu * t), bioreactor volume V(t) = V0 + F * t, compute the total accumulated mass of product M_p(T) from t = 0 to T by definite integration.

Q30. Thermal sterilization of bioprocess media follows first-order thermal death kinetics: dN/dt = -k(T) * N, where k(T) = k0 * e^(-Ea / (R * T(t))). Derive the Del Factor nabla = ln(N0 / Nt) = int_0^t k(T) dt and compute nabla for a linear heating profile T(t) = T0 + alpha * t.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[B]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[B]	**Q22:**	[B]	**Q42:**	[A]	**Q62:**	[B]	**Q82:**	[A]
Q3:	[B]	**Q23:**	[B]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[B]
Q4:	[B]	**Q24:**	[B]	**Q44:**	[C]	**Q64:**	[A]	**Q84:**	[B]
Q5:	[A]	**Q25:**	[B]	**Q45:**	[B]	**Q65:**	[B]	**Q85:**	[C]
Q6:	[B]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[C]
Q7:	[B]	**Q27:**	[B]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[B]	**Q28:**	[B]	**Q48:**	[B]	**Q68:**	[B]	**Q88:**	[A]
Q9:	[B]	**Q29:**	[B]	**Q49:**	[B]	**Q69:**	[B]	**Q89:**	[A]
Q10:	[C]	**Q30:**	[B]	**Q50:**	[C]	**Q70:**	[C]	**Q90:**	[A]
Q11:	[B]	**Q31:**	[B]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[B]	**Q52:**	[B]	**Q72:**	[B]	**Q92:**	[B]
Q13:	[B]	**Q33:**	[B]	**Q53:**	[B]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[B]	**Q34:**	[A]	**Q54:**	[B]	**Q74:**	[B]	**Q94:**	[A]
Q15:	[B]	**Q35:**	[B]	**Q55:**	[B]	**Q75:**	[B]	**Q95:**	[A]
Q16:	[B]	**Q36:**	[B]	**Q56:**	[B]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[B]	**Q37:**	[A]	**Q57:**	[B]	**Q77:**	[B]	**Q97:**	[B]
Q18:	[B]	**Q38:**	[B]	**Q58:**	[C]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[B]	**Q39:**	[B]	**Q59:**	[B]	**Q79:**	[B]	**Q99:**	[A]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[B]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

= -sin^(n-1) x * cos x + (n - 1) * int sin^(n-2) x dx - (n - 1) * int sin^n x dx
= -sin^(n-1) x * cos x + (n - 1) * I_{n-2} - (n - 1) * I_n.
- Bring (n - 1)*I_n to the left-hand side:
I_n + (n - 1)*I_n = -sin^(n-1) x * cos x + (n - 1) * I_{n-2}
n * I_n = -sin^(n-1) x * cos x + (n - 1) * I_{n-2}
I_n = - [sin^(n-1) x * cos x] / n + [(n - 1) / n] * I_{n-2}. (Proved).
2. Deduction of Wallis' Formula over [0, pi/2]:
- Let J_n = int_0^(pi/2) sin^n x dx.
- The boundary term [-sin^(n-1) x * cos x / n]_0^(pi/2) vanishes at both limits:
At pi/2: cos(pi/2) = 0. At 0: sin(0) = 0.
- Therefore: J_n = [(n - 1) / n] * J_{n-2}.
- Case 1: n is an odd positive integer (n = 2k + 1):
J_n = [(n-1)/n] * [(n-3)/(n-2)] * ... * (2/3) * J_1.
J_1 = int_0^(pi/2) sin x dx = [-cos x]_0^(pi/2) = 0 - (-1) = 1.
Wallis formula (odd n): J_n = [(n-1)(n-3)...(2)] / [n(n-2)...(3)] * 1.
- Case 2: n is an even positive integer (n = 2k):
J_n = [(n-1)/n] * [(n-3)/(n-2)] * ... * (1/2) * J_0.
J_0 = int_0^(pi/2) 1 dx = pi / 2.
Wallis formula (even n): J_n = [(n-1)(n-3)...(1)] / [n(n-2)...(2)] * (pi / 2).

>> Solution to Question 8: State and derive the reduction formula for I_n = int cos^n x dx. Using Wallis' formula, evaluate: (i) int_0^(pi/2) cos^7 x dx, and (ii) int_0^(pi/2) cos^

[Problem Statement]: Derive the reduction formula for I_{m,n} = int sin^m x * cos^n x dx. Hence, evaluate the definite integral: int_0^(pi/2) sin^6 x
[Detailed Step-by-Step Answer]:
1. Derivation of Reduction Formula:
- Let I_{m,n} = int sin^m x * cos^n x dx = int [sin^m x * cos^(n-1) x] * cos x dx.
- Integrate by parts: let u = sin^m x * cos^(n-1) x and dv = cos x dx (v = sin x).
du = [m * sin^(m-1) x * cos x * cos^(n-1) x - (n - 1) * sin^m x * cos^(n-2) x * sin x] dx
= [m * sin^(m-1) x * cos^n x - (n - 1) * sin^(m+1) x * cos^(n-2) x] dx.
- I_{m,n} = sin^m x * cos^(n-1) x * sin x - int sin x * du
= sin^(m+1) x * cos^(n-1) x - m * int sin^m x * cos^n x dx + (n - 1) * int sin^(m+2) x *
cos^(n-2) x dx.
- In the last integral, write sin^(m+2) x = sin^m x * (1 - cos^2 x):
int sin^(m+2) x * cos^(n-2) x dx = int sin^m x * cos^(n-2) x dx - int sin^m x * cos^n x dx
= I_{m, n-2} - I_{m, n}.
- Substitute:
I_{m,n} = sin^(m+1) x * cos^(n-1) x - m * I_{m,n} + (n - 1) * I_{m, n-2} - (n - 1) * I_{m,n}
I_{m,n} * [1 + m + n - 1] = sin^(m+1) x * cos^(n-1) x + (n - 1) * I_{m, n-2}
(m + n) * I_{m,n} = sin^(m+1) x * cos^(n-1) x + (n - 1) * I_{m, n-2}
I_{m,n} = [sin^(m+1) x * cos^(n-1) x] / (m + n) + [(n - 1) / (m + n)] * I_{m, n-2}.
(Proved).
2. Evaluation of int_0^(pi/2) sin^6 x * cos^4 x dx:
- Over [0, pi/2], the boundary term vanishes since sin(0) = 0 and cos(pi/2) = 0.
- Wallis' formula for product:
J_{m,n} = [(m-1)(m-3)... * (n-1)(n-3)...] / [(m+n)(m+n-2)...] * K,
where K = pi / 2 if both m and n are even, and K = 1 otherwise.
- Here m = 6 (even), n = 4 (even), so K = pi / 2.
- Numerator:
(m - 1)(m - 3)(m - 5) = (5)(3)(1) = 15.
(n - 1)(n - 3) = (3)(1) = 3.
Numerator product = 15 * 3 = 45.
- Denominator:
(m + n)(m + n - 2)... = (10)(8)(6)(4)(2) = 3840.
- Result:
J_{6,4} = [45 / 3840] * (pi / 2) = [3 / 256] * (pi / 2) = 3*pi / 512.
Conclusion:
int_0^(pi/2) sin^6 x * cos^4 x dx = 3*pi / 512.

>> Solution Strategy: Derive the reduction formulas for: (i) I_n = int tan^n x dx, and (ii) I_n = int sec^n x dx. Using these, evaluate int_0^(pi/4) tan^5 x dx and i

2. Part (ii): Reduction Formula for I_n = int sec^n x dx:
- Write sec^n x = sec^(n-2) x * sec^2 x.
- Integrate by parts: let u = sec^(n-2) x, dv = sec^2 x dx (v = tan x).
du = (n - 2) * sec^(n-3) x * (sec x * tan x) dx = (n - 2) * sec^(n-2) x * tan x dx.
I_n = sec^(n-2) x * tan x - (n - 2) * int sec^(n-2) x * tan^2 x dx.
- Use tan^2 x = sec^2 x - 1:
I_n = sec^(n-2) x * tan x - (n - 2) * [I_n - I_{n-2}]
I_n + (n - 2)*I_n = sec^(n-2) x * tan x + (n - 2) * I_{n-2}
(n - 1)*I_n = sec^(n-2) x * tan x + (n - 2) * I_{n-2}
I_n = [sec^(n-2) x * tan x] / (n - 1) + [(n - 2) / (n - 1)] * I_{n-2}. (Proved).
- Evaluate I_3 = int sec^3 x dx:
Set n = 3:
I_3 = [sec x * tan x] / 2 + (1 / 2) * I_1.
Since I_1 = int sec x dx = ln|sec x + tan x|:
I_3 = (1/2) * sec x * tan x + (1/2) * ln|sec x + tan x| + C.

>>Solution Question 6: Derive the reduction formula for I_n = int x^n e^(ax) dx. Use this formula to evaluate the definite integral int_0^1 x^4 e^(2x) dx.

[Detailed Step-by-Step Answer]:
1. Derivation of Reduction Formula:
- Let I_{m,n} = int x^m * (ln x)^n dx.
- Integrate by parts: let u = (ln x)^n, dv = x^m dx.
du = n * (ln x)^(n-1) * (1 / x) dx, v = x^(m+1) / (m + 1).
- I_{m,n} = [x^(m+1) * (ln x)^n] / (m + 1) - int [x^(m+1) / (m + 1)] * [n * (ln x)^(n-1) / x] dx
= [x^(m+1) * (ln x)^n] / (m + 1) - [n / (m + 1)] * int x^m * (ln x)^(n-1) dx
= [x^(m+1) * (ln x)^n] / (m + 1) - [n / (m + 1)] * I_{m, n-1}. (Proved).
2. Definite Integral over [0, 1]:
- Boundary term: [x^(m+1) * (ln x)^n / (m + 1)]_0^1:
At x = 1: ln(1) = 0 => 0.
At x -> 0+: By L'Hopital's rule, lim_{x -> 0+} x^(m+1) * (ln x)^n = 0 (for m + 1 > 0).
Thus, the boundary term vanishes completely at both limits.
- Recurrence relation for definite integral J_{m,n} = int_0^1 x^m * (ln x)^n dx:
J_{m,n} = - [n / (m + 1)] * J_{m, n-1}.
- Repeated application gives:
J_{m,n} = (-1)^n * [n! / (m + 1)^n] * J_{m, 0}.
- Base integral J_{m, 0}:
J_{m, 0} = int_0^1 x^m dx = [x^(m+1) / (m + 1)]_0^1 = 1 / (m + 1).
- General formula:
int_0^1 x^m * (ln x)^n dx = (-1)^n * [n! / (m + 1)^(n + 1)].
3. Evaluation for m = 3, n = 2:
- Substitute m = 3, n = 2 into the general formula:
int_0^1 x^3 * (ln x)^2 dx = (-1)^2 * [2! / (3 + 1)^(2 + 1)]
= 1 * [2 / 4^3]
= 2 / 64 = 1 / 32.
Conclusion:
int_0^1 x^3 * (ln x)^2 dx = 1 / 32.

>> Solution to Question 8: [Problem Statement]: Define the Gamma function Gamma(n), Prove that the integral converges for all n > 0, Establish the recurrence relation Gamma(n + 1)

- Integrate by parts: let u = x^n, dv = e^(-x) dx (v = -e^(-x)):
du = n * x^(n - 1) dx.
- Gamma(n + 1) = [-x^n * e^(-x)]_0^infinity - int_0^infinity (-e^(-x)) * [n * x^(n - 1)] dx
= [0 - 0] + n * int_0^infinity e^(-x) * x^(n - 1) dx
= n * Gamma(n). (Proved).
3. Deduction that Gamma(n + 1) = n!:
- Evaluate Gamma(1):
Gamma(1) = int_0^infinity e^(-x) * x^0 dx = int_0^infinity e^(-x) dx = [-e^(-x)]_0^infinity = 0
- (-1) = 1.
- By repeated application of recurrence:
Gamma(2) = 1 * Gamma(1) = 1 = 1!
Gamma(3) = 2 * Gamma(2) = 2 * 1! = 2!
Gamma(4) = 3 * Gamma(3) = 3 * 2! = 3!
In general, for any integer n >= 0:
Gamma(n + 1) = n * (n - 1) * ... * 1 * Gamma(1) = n!.

>> Solution to Question 30: Prove that Gamma(1/2) = sqrt(pi) by transforming the product of two Gaussian integrals into polar coordinates.

[Problem Statement]: Define the Beta function Beta(m, n). Prove: (i) Beta(m, n) = Beta(n, m) (symmetry), and (ii) Beta(m, n) = 2 * int_0^(pi/2) sin^
[Detailed Step-by-Step Answer]:
1. Definition:
The Beta function (Euler's integral of the first kind) is defined for m > 0, n > 0 by:
Beta(m, n) = int_0^1 x^(m - 1) * (1 - x)^(n - 1) dx.
2. Proof of Symmetry: Beta(m, n) = Beta(n, m):
- In Beta(m, n) = int_0^1 x^(m - 1) * (1 - x)^(n - 1) dx, substitute x = 1 - t:
dx = -dt.
Limits: when x = 0, t = 1; when x = 1, t = 0.
- Beta(m, n) = int_1^0 (1 - t)^(m - 1) * t^(n - 1) * (-dt)
= int_0^1 t^(n - 1) * (1 - t)^(m - 1) dt.
- Since t is a dummy variable, this is identically Beta(n, m):
Beta(m, n) = Beta(n, m). (Proved).
3. Proof of Trigonometric Form:
- In Beta(m, n) = int_0^1 x^(m - 1) * (1 - x)^(n - 1) dx, substitute x = sin^2 theta:
dx = 2 * sin theta * cos theta dtheta.
1 - x = 1 - sin^2 theta = cos^2 theta.
Limits: when x = 0, theta = 0; when x = 1, theta = pi / 2.
- Substitute into integral:
Beta(m, n) = int_0^(pi/2) (sin^2 theta)^(m - 1) * (cos^2 theta)^(n - 1) * (2 * sin theta * cos
theta) dtheta
= int_0^(pi/2) sin^(2m - 2) theta * cos^(2n - 2) theta * 2 * sin theta * cos theta dtheta
= 2 * int_0^(pi/2) sin^(2m - 1) theta * cos^(2n - 1) theta dtheta. (Proved).
4. Direct Corollary for Wallis' Integrals:
Dividing by 2 gives:
int_0^(pi/2) sin^p theta * cos^q theta dtheta = (1/2) * Beta((p + 1)/2, (q + 1)/2).

>>Solution Question 11: [Problem Statement]: Derive the fundamental relation connecting Beta and Gamma functions: Beta(m, n) = [Gamma(m) * Gamma(n)] / Gamma(m + n).

- Factor 1: 2 * int_0^(pi/2) sin^(2m - 1) theta * cos^(2n - 1) theta dtheta = Beta(m, n) (Trigonometric form).
- Factor 2: In 2 * int_0^infinity e^(-r^2) * r^(2(m+n) - 1) dr, substitute u = r^2 (du = 2r dr):

int_0^infinity e^(-u) * u^(m + n - 1) du = Gamma(m + n).

Step 6: Combine:

Gamma(m) * Gamma(n) = Beta(m, n) * Gamma(m + n).

Divide both sides by Gamma(m + n):

Beta(m, n) = [Gamma(m) * Gamma(n)] / Gamma(m + n).

Hence proved.

For any non-integer p in (0, 1):
Gamma(p) * Gamma(1 - p) = pi / sin(p * pi).
2. Part (i): Evaluate Gamma(1/4) * Gamma(3/4):
- Set p = 1/4 (so 1 - p = 3/4):
Gamma(1/4) * Gamma(3/4) = pi / sin(pi / 4).
- Since sin(pi / 4) = 1 / sqrt(2):
Gamma(1/4) * Gamma(3/4) = pi / (1 / sqrt(2)) = pi * sqrt(2).
3. Part (ii): Evaluate I = int_0^infinity [1 / (1 + x^4)] dx:
- Substitute x^4 = t => x = t^(1/4) => dx = (1/4) * t^(-3/4) dt.
Limits: when x = 0, t = 0; when x -> infinity, t -> infinity.
- I = int_0^infinity [1 / (1 + t)] * (1/4) * t^(-3/4) dt
= (1/4) * int_0^infinity [t^(1/4 - 1) / (1 + t)] dt.
- Use the standard Beta function identity:
int_0^infinity [t^(p - 1) / (1 + t)] dt = Beta(p, 1 - p) = Gamma(p) * Gamma(1 - p).
- Here p = 1/4:
I = (1/4) * Beta(1/4, 3/4) = (1/4) * [Gamma(1/4) * Gamma(3/4) / Gamma(1)].
- Using Gamma(1) = 1 and Gamma(1/4)*Gamma(3/4) = pi * sqrt(2):
I = (1/4) * pi * sqrt(2) = pi / (2 * sqrt(2)).
Conclusion:
(i) Gamma(1/4) * Gamma(3/4) = pi * sqrt(2).
(ii) int_0^infinity [1 / (1 + x^4)] dx = pi / (2 * sqrt(2)).

Solution to Question 34: Evaluate the definite integrals using Beta and Gamma functions: (i) int_0^1 x^4 * (1 - sqrt(x))^5 dx, and (ii) int_0^(pi/2) sqrt(tan theta) dt.

>>Solution Statement: We have to prove that int_0^(pi/2) sin x/(1+cos^2 x) dx = pi/4. Solution: We know properties of definite integrals.

=> I = pi^2 / 4.
Hence proved.

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ds^2 = 2*a^2 * [2 * sin^2(theta / 2)] = 4*a^2 * sin^2(theta / 2).
Taking square root (since theta in [0, 2*pi], theta/2 in [0, pi], so sin(theta/2) >= 0):
ds = 2a * sin(theta / 2) dtheta.
Step 4: Integrate from theta = 0 to theta = 2*pi:
s = int_0^(2*pi) 2a * sin(theta / 2) dtheta
= 2a * [- 2 * cos(theta / 2)]_0^(2*pi)
= -4a * [cos(pi) - cos(0)]
= -4a * [(-1) - (1)]
= -4a * (-2) = 8a.
Conclusion:
The arc length of one complete arch of the cycloid is exactly 8a.

>>**Solution Statement Question 28:** The area bounded by the curve y = cosh(x/a), between x = -a and x = a, is ds^2 = (a dtheta)^2 dtheta. Hence, compute the total perimeter of the cardio

Step 2: Compute 1 + (dy/dx)^2:
1 + (dy/dx)^2 = 1 + sinh^2(x / c).
Using the fundamental hyperbolic identity cosh^2(u) - sinh^2(u) = 1:
1 + sinh^2(x / c) = cosh^2(x / c).
Step 3: Square root for arc length element:
ds = sqrt(cosh^2(x / c)) dx = cosh(x / c) dx (since cosh(u) >= 1 > 0 for all real u).
Step 4: Integrate from x = -a to x = a:
By symmetry of the even function cosh(x / c):
s = int_{-a}^a cosh(x / c) dx = 2 * int_0^a cosh(x / c) dx.
Step 5: Perform the integration:
s = 2 * [c * sinh(x / c)]_0^a
= 2c * [sinh(a / c) - sinh(0)]
= 2c * sinh(a / c).
Conclusion:
The total arc length of the catenary is 2c * sinh(a / c).

>>Solution Step Question 22: The area between r = 1 + cos(θ) and r = 1 − cos(θ) is the area of the entire region enclosed by the cardio

[Detailed Step-by-Step Answer]:
1. Derivation of Polar Area Formula:
- In polar coordinates, consider an infinitesimal sector swept by radius vector r through an angle dtheta.
- The sector approximates a circular sector of radius r and central angle dtheta:
dA = (1/2) * r * (r * dtheta) = (1/2) * r^2 dtheta.
- Total area from theta = alpha to theta = beta is obtained by summing these differential sectors:
A = (1/2) * int_alpha^beta r^2 dtheta. (Proved).
2. Computation for Cardioid r = a * (1 + cos theta):
- By symmetry about the initial line theta = 0, the total area is:
A = 2 * [(1/2) * int_0^pi r^2 dtheta] = int_0^pi r^2 dtheta.
- Substitute r = a * (1 + cos theta):
r^2 = a^2 * (1 + cos theta)^2 = a^2 * (1 + 2*cos theta + cos^2 theta).
- Use double-angle formula cos^2 theta = (1 + cos 2theta) / 2:
r^2 = a^2 * [1 + 2*cos theta + 1/2 + (1/2)*cos 2theta]
= a^2 * [3/2 + 2*cos theta + (1/2)*cos 2theta].
- Integrate from 0 to pi:
A = a^2 * int_0^pi [3/2 + 2*cos theta + (1/2)*cos 2theta] dtheta
= a^2 * [(3/2)*theta + 2*sin theta + (1/4)*sin 2theta]_0^pi.
- Evaluate at limits:
At pi: (3/2)*pi + 2*sin(pi) + (1/4)*sin(2pi) = (3/2)*pi + 0 + 0 = 3*pi / 2.
At 0: 0.
A = a^2 * (3*pi / 2) = (3/2) * pi * a^2.
Conclusion:
The area enclosed by the cardioid is (3/2) * pi * a^2.

>>**Solution to Question 23:** Derive the Disk Method for the volume of a solid of revolution. Use it to prove that the volume of a sphere of radius R is (4/3) * pi * R^3.

>>**Solution Statement Question 25:** Derive the formula for the surface area of a solid of revolution about the x-axis. S = 2*pi*int_a^b f(x)*sqrt(1 + (f'(x))^2) dx. Use this method to find the

- Integrate from -R to R:
S = 2 * pi * int_{-R}^R R dx = 2 * pi * R * [x]_{-R}^R
= 2 * pi * R * [R - (-R)] = 2 * pi * R * (2R)
= 4 * pi * R^2.
Hence proved.

- Integrate P2(t) from -h to h:
int_{-h}^h (A*t^2 + B*t + C) dt = [A*t^3 / 3 + B*t^2 / 2 + C*t]_{-h}^h
= (2/3)*A*h^3 + 2*C*h = (h / 3) * [2*A*h^2 + 6*C].
- Substitute 2*A*h^2 = y0 + y2 - 2*y1 and C = y1:
int_{x0}^{x2} f(x) dx ~ (h / 3) * [(y0 + y2 - 2*y1) + 6*y1] = (h / 3) * [y0 + 4*y1 + y2].
- Summing over all n/2 double-subintervals gives the Composite Simpson's 1/3 Rule:
int_a^b f(x) dx ~ (h / 3) * [(y0 + yn) + 4*(y1 + y3 + ... + y_{n-1}) + 2*(y2 + y4 + ... + y_{n-2})].
2. Error Term and Degree of Precision:
- By Taylor series expansion, the cubic term cancels due to symmetry!
- Local error for one pair of intervals: E_local = - (h^5 / 90) * f^(4)(xi).
- Total global error:
E_global = - [(b - a) * h^4 / 180] * f^(4)(eta) = O(h^4).
- Simpson's 1/3 rule has degree of precision 3: it integrates polynomials up to degree 3 (cubic) exactly.

X(t) = X0 * e^(mu * t)
V(t) = V0 + F * t.
- Total product mass accumulated at time T:
M_p(T) = int_0^T q_p * X0 * e^(mu * t) * (V0 + F * t) dt
= q_p * X0 * [V0 * int_0^T e^(mu * t) dt + F * int_0^T t * e^(mu * t) dt].
2. Evaluate Integral 1:
I1 = int_0^T e^(mu * t) dt = [(1 / mu) * e^(mu * t)]_0^T = (1 / mu) * (e^(mu * T) - 1).
3. Evaluate Integral 2 (Integration by Parts):
I2 = int_0^T t * e^(mu * t) dt.
Let u = t, dv = e^(mu * t) dt => du = dt, v = (1 / mu) * e^(mu * t).
I2 = [(t / mu) * e^(mu * t)]_0^T - (1 / mu) * int_0^T e^(mu * t) dt
= (T / mu) * e^(mu * T) - (1 / mu^2) * (e^(mu * T) - 1).
4. Combine Terms:
M_p(T) = q_p * X0 * [(V0 / mu) * (e^(mu * T) - 1) + (F * T / mu) * e^(mu * T) - (F / mu^2) * (e^(mu * T) - 1)]
= (q_p * X0 / mu) * [(V0 - F / mu) * (e^(mu * T) - 1) + F * T * e^(mu * T)].
5. Analytical Verification:
When F = 0 (constant volume batch reactor):
M_p(T) = (q_p * X0 * V0 / mu) * (e^(mu * T) - 1), which matches standard batch product accumulation.

Solution to Question 30: Thermal sterilization of bioprocess media follows first-order thermal death kinetics: dN/dt = -k(T) * N, where k(T) = k0 * e^(-Ea / (R * T(t)